

Comparative Study of Laser Doppler Vibrometer and Capacitive Air-coupled Transducer for Ultrasonic Propagation Imager and the New Development of an Efficient Ultrasonic Wavenumber Imaging Algorithm

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ABSTRACT: Damage detection techniques using guided waves have been studied for decades with very few successful real-world applications. The recent development with the full wavefield technique using the Ultrasonic Propagation Imager (UPI) is one of those few exceptions. In this paper, we study two non-contact sensors: the laser Doppler vibrometer and the capacitive air-coupled transducer in the context as the sensing modules for the UPI. The aim of this paper is to provide a comprehensive study for optimisation of the two sensors, as well as a comparison between them for use in the UPI. First, the parameters for laser ultrasonic measurement of each sensor were studied: surface treatment, measurement angle and stand-off distance in the case of the laser Doppler vibrometer and measurement angle, lift-off distance and bias voltage in the case of the capacitive air-coupled transducer. Two optimised sensors were then compared in terms of their ability to detect damages using the UPI. Also, in this paper, we presented the ultrasonic wavenumber imaging (UWI) algorithm with the new development towards an efficient implementation. The uniqueness of the UWI algorithm with the capability of damage size estimation makes this algorithm very attractive for the future study with full wavefield signal processing.

KEY WORDS: *capacitive air-coupled transducer, laser Doppler vibrometer, laser Ultrasonic Propagation Imager, performance comparison, ultrasonic wavenumber imaging*

Introduction

Damage detection techniques using guided waves have been studied for decades with very few successful real-world applications [1]. The recent development with the full wavefield technique using the Ultrasonic Propagation Imager (UPI), which includes a Q-switched laser and a two-dimensional mirror scanner accompanied with the sensing and data acquisition systems, is one of those few exceptions. The UPI overcomes two basic disadvantages of measurement techniques using a sparse sensor network: (1) the information obtained with few sensors is often too limited to clearly detect the existence, location and severity of damages, especially for real-world structures with fairly complicated features such as spar, rivets, reinforcement, etc. and (2) damage detection techniques using a sparse sensor network often require a baseline data, which are the measurement data of the structure before damage. The requirement of comparison with baseline data is not only impractical in the real world but also extremely susceptible to environmental and operational variations such as temperature changes and ambient forces during measurement [2]. The full wavefield techniques provide much richer information with high measurement resolution, and also baseline data are not needed for the purpose of damage detection.

The UPI is one among several full wavefield techniques. The term 'full wavefield' refers to the set of data

(displacement, velocity, etc.) of dense grid points in an interest area measured during a time duration. Other techniques that enable the measurement of full wavefield are scanning air-coupled transducer (ACT) and scanning laser vibrometry [3]. The scanning ACT uses a piezoelectric transducer for excitation at a fixed location and an ACT moving at various locations in the scan area to measure the ultrasonic waves [4]. The scanning laser vibrometry technique operates with the same principle, where a continuous laser beam is used to measure the ultrasonic waves at various locations in the scan area using a scan mirror [5]. The advantage of the scanning laser vibrometry technique is that it eliminates the bulky mechanical part to move the ACT. However, this technique requires a target with a smooth and reflective surface, which is often not the case with real-world structures. One solution is to use retro-reflective tape and attach it to each and every measurement point, which makes it extremely laborious and time-consuming to properly prepare the test. The UPI [6] has a different approach to obtain the full wavefield. Instead of using one fixed excitation source and moving the sensing laser at various points, this technique uses one or many fixed sensing points and controls a Q-switched laser beam for scanning (see Figure 1). Through reciprocity, a virtual laser ultrasonic wavefield is generated from the sensing point(s). The advantages of this technique are as follows: (1) it is applicable to realistic surfaces, in the sense that there is no need to prepare the structure's surface, (2) it allows for

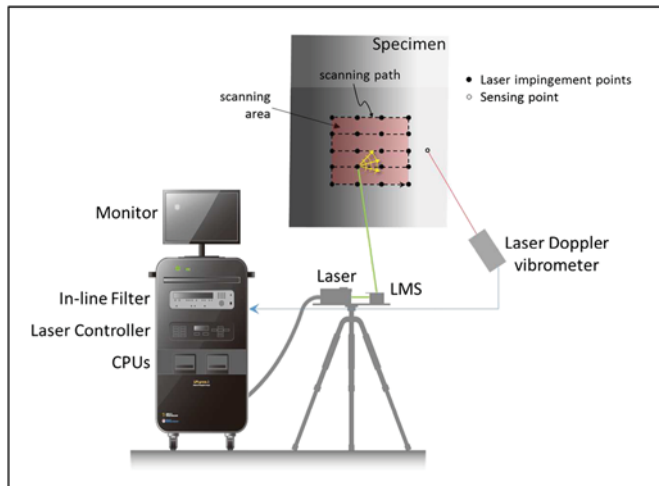


Figure 1: Experimental set-up with Ultrasonic Propagation Imager (UPI) (LMS: laser mirror scanner, CPU: central processing unit)

the manufacture of very-high-speed scanning systems—thanks to the availability of high-speed Q-switching to generate laser pulse, high-speed scan mirror and high-speed data acquisition. The latest generation of the Q-switched laser scanning system [7] has reached a scan speed of 20 000 points per second, which means that the inspection time for an area of $1\text{ m} \times 1\text{ m}$ with a spatial resolution of 1 mm is less than 1 min , (3) The Q-switched laser scanning system is not fixed to any particular sensing system but can be hooked up to a wide range of sensor types, and (4) it is easy to extend the inspected area by using a set of electric-controlled mirrors to alter the direction of the laser beam [8].

However, there are several disadvantages of the UPI [9]. First, the excitation type is often limited as single pulse excitation. And second, the focus of a high-energy laser beam in an ultra-short period of time may damage the inspected specimen. Current research is ongoing to address these issues, either by using spatial modulation technique, which involves spreading the laser energy in space domain using lenticular lens [10], optical diffraction gratings [11], transmission mask [12] and laser-induced grating [13], or by using temporal modulation technique, which involves spreading the laser energy in the time domain using Bragg cell [14], pulsed or continuous-wave laser Q-switching technique [6, 15] and intensity modulation of diode laser [16].

In this paper, we detail our study of two non-contact sensors: the laser Doppler vibrometer (LDV) and the capacitive ACT as sensing modules for the laser UPI. The aim of this paper is to provide a comprehensive study for optimising the two sensors, as well as to provide a comparison between them for use in the UPI system. However, it must be pointed out that the comparison presented here is only relative. As an example, the sensing

distance of the ACT is in the centimetre scale, whereas for the LDV, the sensing distance is in metres.

The paper is organised as follows. We start with a brief introduction of the working principle of the capacitive ACT and the LDV. Next, each sensor was optimised with various parameters for laser ultrasonic measurement: measurement angle, lift-off distance and bias voltage for the capacitive ACT and surface treatment, measurement angle and stand-off distance for the LDV. Two optimised sensors were then compared in terms of their ability to detect damages using the UPI. Also, for the purpose of performance comparison, we presented the ultrasonic wavenumber imaging (UWI) algorithm including new development towards an efficient implementation. The paper concludes with a summarised comparison between LDV and the capacitive ACT systems for use in the UPI.

Parameter Study of the Capacitive Air-coupled Transducer and the Laser Doppler Vibrometer for the Ultrasonic Propagation Imager System

In this section, we present the working principles of the capacitive ACT and the LDV to determine the measurement parameters most critical for laser ultrasonic measurement. An experiment with a single-point laser excitation and sensing was set up to verify those working principles as shown in Figure 2. The specimen is a 5-mm -thick aluminium plate. The laser pulse is excited at a distance of 50 mm from the sensing point.

Capacitive air-coupled transducer

The laser ultrasonic wave travels through the plate and leaks into the air across the solid/air interface. The angle θ between the leaky wave and the normal vector of the plate is given by the well-known Snell law:

$$\theta = \sin^{-1}(v_a/v_w) \quad (1)$$

where v_a and v_w represent the velocity of the wave in air and in the plate, respectively. Since the ultrasonic wave travels in the plate at different modes corresponding to different wave velocities, choosing the appropriate measurement angle enables the ability to selectively measure a distinct mode of the ultrasonic wave. For a structure with known thickness and material, the measurement angle can be easily calculated using the theoretical dispersion curve shown in Figure 3. Otherwise, the velocity of the interest mode can be measured experimentally.

The second important parameter of the ACT is the lift-off distance since the ultrasonic wave travelling through air is strongly attenuated [17]. The attenuation can be seen as a reduction of peak-to-peak amplitude of the wave, which decays exponentially with distance given by

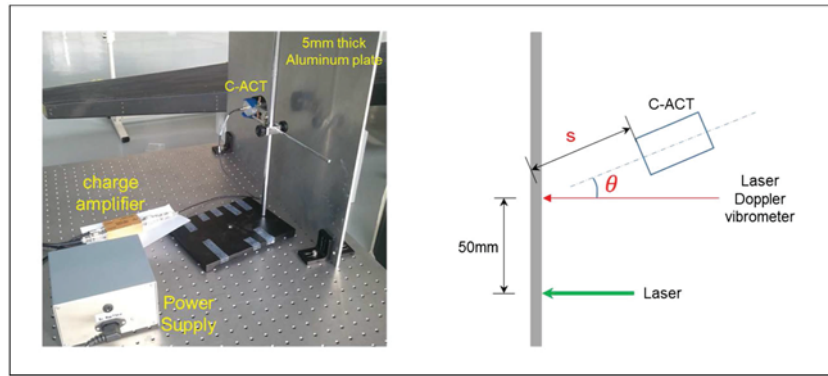


Figure 2: Experimental set-up to study the working principles of the capacitive air-coupled transducer (ACT) and the laser Doppler vibrometer

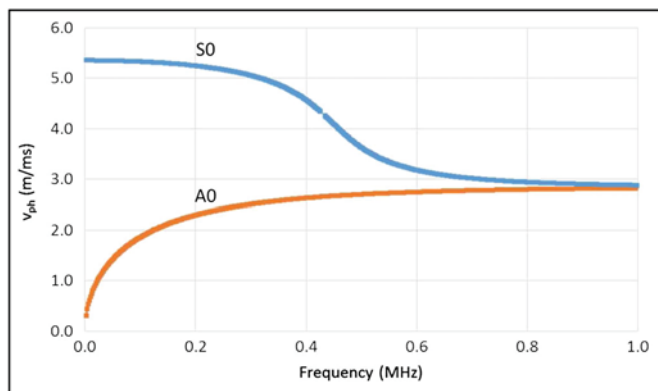


Figure 3: Theoretical dispersion curve of the 5-mm-thick aluminium plate

$$p = p_0 \exp(-\alpha s) \quad (2)$$

where p is the peak amplitude of the wave at the propagation distance s , p_0 is the initial peak amplitude and α is the attenuation coefficient. Figure 4 shows the ultrasonic peak-to-peak amplitude plotted against the lift-off distance as measured with the ACT. The experimental data are well fitted by an exponential curve with the attenuation coefficient of the air measuring approximately 0.0112 Np m^{-1} .

The detailed design of the ACT as shown in Figure 5 reveals another parameter that strongly affects the measurement of the ACT: the bias voltage applied between the solid back plate and the polymer membrane. During measurement, the polymer membrane moves due to fluctuations of the air pressure. This causes a change in the separation between the two parallel plates, and consequently, a change in the capacitance $\Delta C = -\epsilon_0 A \Delta x / d^2$. Hence, the charge difference between the two plates is

$$\Delta Q = -\frac{\epsilon_0 A \Delta x}{d^2} V_{DC} \quad (3)$$

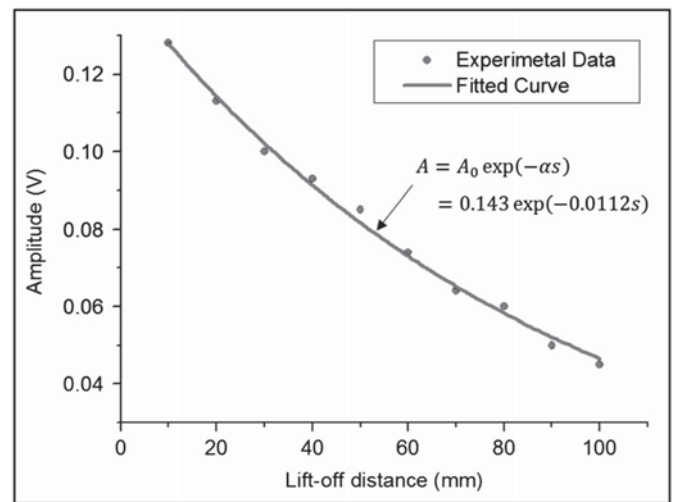


Figure 4: Laser ultrasonic peak-to-peak amplitude versus lift-off distance measured using the capacitive air-coupled transducer

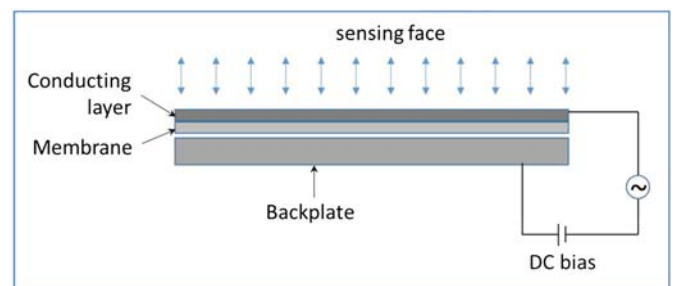


Figure 5: Configuration of a capacitive air-coupled transducer

where V_{DC} is the bias voltage, A is the surface area, d is the thickness of the air gap, ϵ_0 is the permittivity of free space and Δx is the change of separation between two plates. Experimental results shown in Figure 6 confirm that the measurement output of the ACT is proportional to the DC bias as presented in Equation [3].

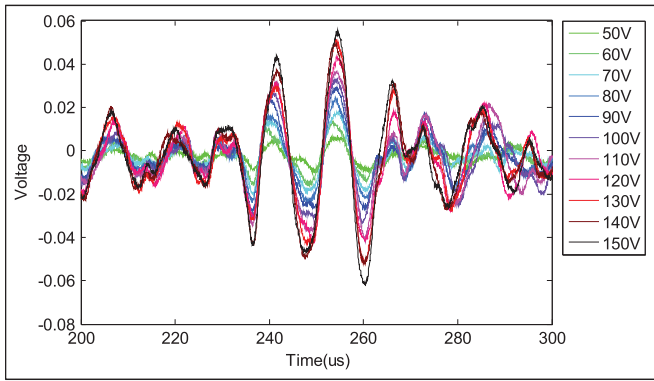


Figure 6: Laser ultrasonic signal at various applied bias voltages measured using the capacitive air-coupled transducer

Laser Doppler vibrometer

An LDV measures the velocity of an object by detecting the Doppler frequency shift that the vibrating structure imposes on the impinging laser beam [18]. The optical configuration of the LDV is shown in Figure 7. The laser beam is split into reference and measurement beams by a polarising beam splitter (PBS). An acousto-optic modulator (Bragg cell) is incorporated to shift the frequency of the reference beam. The measurement beam passes through a PBS as well as a quarter-wave plate and focuses on the object. The scattered light reflected from the object is collected and combined with the shifted-frequency reference beam at the photodetectors. Further signal processing with modulated signal from the photodetectors provides the velocity of the object.

The ability to collect reflected light from the target plays a central role in the LDV measurement. To get the highest possible signal-to-noise ratio of the measurement signal, the LDV must be optimised to gather the most light from the target. Figure 8a shows the effect of measurement angle at various surface conditions on the signal level. The signal level is maximum when the laser beam is perpendicular to the surface's target. Note that this is true for all three investigated surface conditions. When the aluminium plate is polished, the highly reflective surface allows a relatively

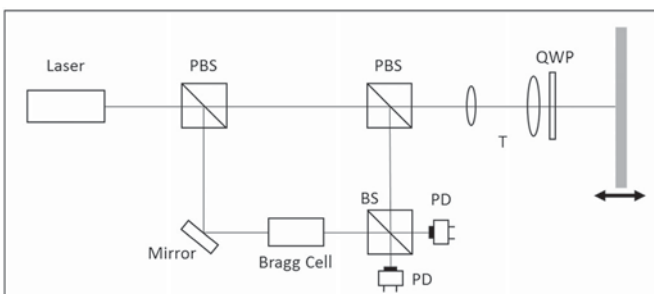


Figure 7: Optical arrangements of the laser Doppler vibrometer (BS: beam splitter, PBS: polarising beam splitter, QWP: quarter-wave plate, PD: photo-detector, T: telescopic lens array)

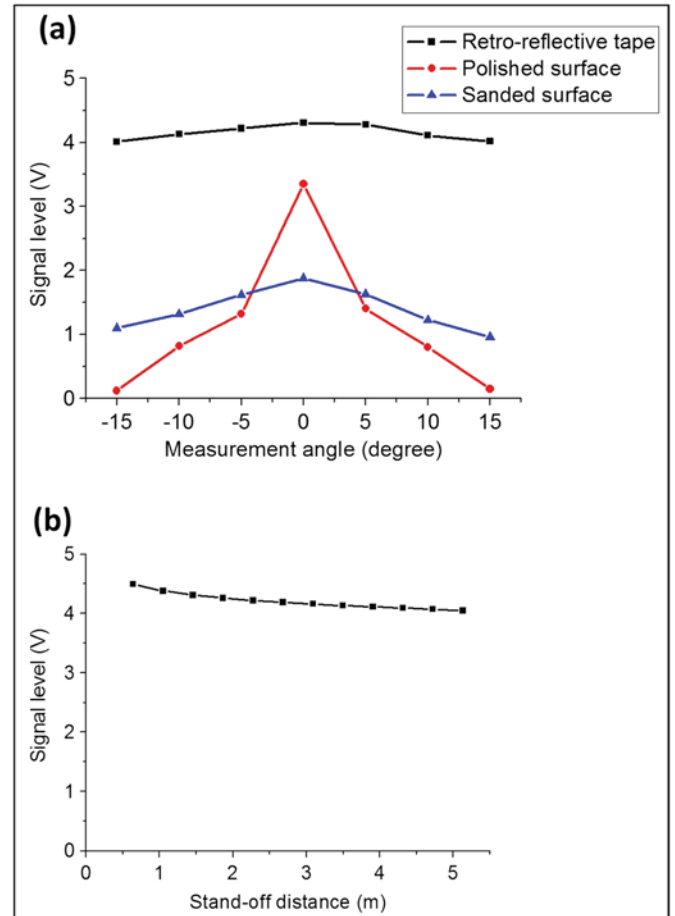


Figure 8: Measurement of signal level of the laser Doppler vibrometer (a) at different measurement angles and surface conditions and (b) at different stand-off distances

high level of reflection. However, it is very sensitive to misalignment of the laser beam: the signal level drops deeply even at a very small misalignment angle. Simply making the surface of the aluminium plate roughly by using sand paper can help to reduce the sensitivity of the signal level to the misalignment, however with the cost of reflection loss at the normal angle. Finally, surface treatment with the retro-reflective tape shows very high signal levels, even with large measurement angles of up to $\pm 15^\circ$.

The third parameter that affects the signal level is the stand-off distance, which is the distance from the LDV to the target. Using the retro-reflective tape and measuring at normal angles, the signal level decreased by less than 12% as the stand-off distance increased up to 5 m (Figure 8b).

Experiment Set-up for Damage Detection

Two experiments were set up to compare the performance of the ACT and the LDV for damage detection using the UPI system. The optimised measurement parameters of each sensor as determined in the previous section were applied.

For the ACT, the optimum measurement angle was calculated using Equation [1]. The lift-off distance was chosen at 30 mm, and the bias voltage applied to the ACT was 150 V. For the LDV, a retro-reflective tape was used, and the beam was perpendicular to the specimen. The stand-off distance was 3.3 m.

The first experiment was set up to detect the impact damages of a 2-mm-thick carbon fibre-reinforced polymer (CFRP) specimen as shown in Figure 9a and b. Figure 9c shows the Q-switched laser and the scanning mirror of the UPI. Figure 9d shows the picture of the laser vibrometers. In this paper, only one LDV in the middle was used for measurement. The second experiment was set up to detect a hole inside a plastic specimen, which was made by a 3D printer as shown in Figure 10. The thickness of the plastic specimen was 8 mm, and the thickness of the hole was 2 mm. The scan areas of experiments 1 and 2 were 100 mm × 100 mm and 50 mm × 50 mm, respectively, both with 0.5-mm interval of the laser scanning points. The laser was set at a pulse rate of 1 kHz, and the energy per pulse was 1.20 mJ. The sensing point for both experiments was at a distance of 10 mm to the edge of the scanning area. The collected laser ultrasonic signal from the sensor was amplified 25 times and passed through a bandpass filter device in the range of 30–300 kHz for both experiments. Since the signal-to-noise ratio (SNR) of non-contact sensing is generally low, it was necessary to use a repeat scanning technique [19] to improve the SNR by precisely controlling the scan mirror to repeat identical scan patterns multiple times and averaging them. For the LDV, we used 15 times

of averaging, and for the ACT, we used five times of averaging. A comparison of ultrasonic signals using two sensors is shown in Figure 11 at the same laser excitation point: at the middle of the lower edge of the scan area of the CFRP wing box specimen. As seen in Figure 11, the peak of the signal from the ACT sensor is delayed approximately 50 μs compared to the signal from the LDV sensor due to the propagation time of the waves in the air. The SNR of a signal is calculated as

$$\text{SNR} = 20 \log \left(\frac{A}{\sigma} \right) \quad (4)$$

where A is the peak-to-peak value of the signal and σ is the standard deviation of the noise. Assuming the first 10 μs of the measurement signals is noise, the SNR of the signals are then calculated using Equation [4]: the SNR of the signal measured with LDV sensor is 18 dB, and the SNR of the signal measured with ACT sensor is 30 dB.

The full scanning data form a three-dimensional matrix $V \times H \times N$, where V and H are the number of scanning points in vertical and horizontal directions of the scanning area and N is the sampling length of measurement data at each point. A slide that is perpendicular to the N axis maps the ultrasonic wave propagation at one particular time point. Figure 12a and b shows the ultrasonic wave propagation images for experiment 1 using the ACT and the LDV sensors, respectively. The double-impact damages were detected in both cases, though the ACT sensor showed the damages more clearly due to its higher SNR. Similar

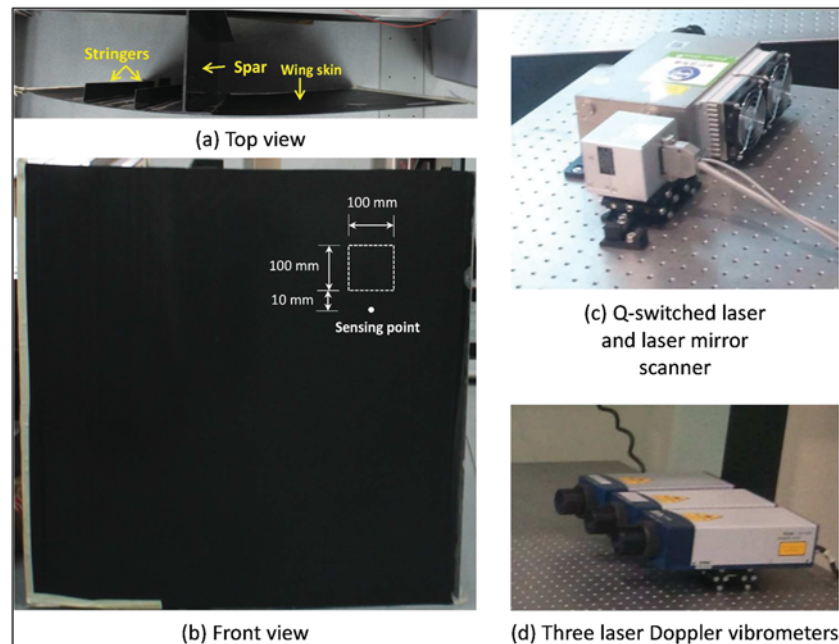


Figure 9: Carbon fibre-reinforced polymer wing box specimen: (a) top view and (b) front view. (c) Q-switched laser and the scan mirror and (d) the laser Doppler vibrometers

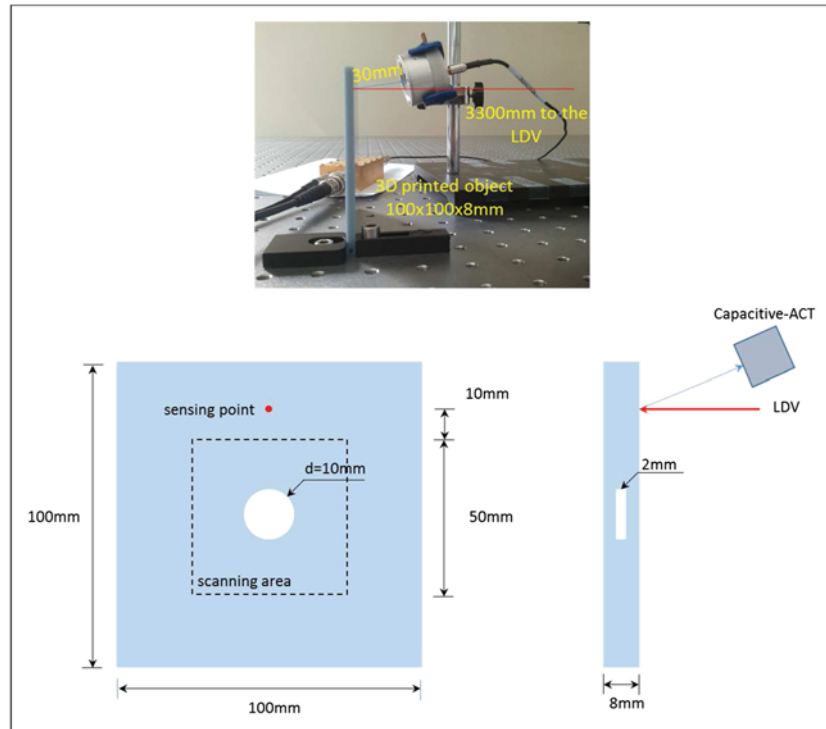


Figure 10: The 3D-printed object 100 mm × 100 mm × 8 mm with a circular hole of 10-mm diameter and 2 mm thick inside the object. ACT, air-coupled transducer; LDV, laser Doppler vibrometer

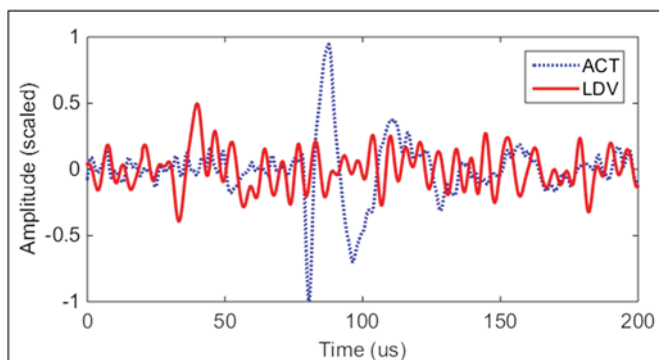


Figure 11: Laser ultrasonic signals measured using air-coupled transducer (ACT) and laser Doppler vibrometer (LDV) sensors

conclusions can be drawn for the second experiment with the 3D-printed object as shown in Figure 13a and b. In the next section, we will explore further the performance comparison in terms of damage detection using the UWI algorithm.

Damage Detection Using Ultrasonic Wavenumber Imaging Algorithm

As shown in the previous section, the damages can be detected using the wave propagation images. By observing the propagation of the ultrasonic waves, a technician can

detect the damage locations where the waves show abnormal amplitude and direction. This technique is often adequate for a preliminary assessment since it requires very little computation and the result is real time. However, there are two major challenges of this simple technique. First, from the machine learning perspective, it is difficult for the recognition of the damages since it is required to analyse multiple frames of the wave propagation to make a decision. Second, in the case when the object has a complex internal structure, such as a honeycomb sandwich plate with a thick aluminium core inside, the damages are barely recognised if using merely the wave propagation images [20].

Recent advances in signal processing of full-wavefield data tackle those challenges and advance the capability of damage detection and size estimation. In this paper, we applied the wavenumber imaging algorithms presented by Flynn *et al.* [21], Rogge *et al.* [22] and Mesnil *et al.* [23]. For the sake of completeness, the mathematical development of the UWI algorithm is presented here. Readers who are interested in a more rigorous mathematical treatment of the wavenumber imaging algorithm can refer to a prior work of Flynn *et al.* [21]. Since this algorithm is computationally intensive, the study to optimise its implementation will be introduced in the next section. Finally, the analysis results using the UWI algorithm for the experiment with the CFRP specimen will be shown and compared.

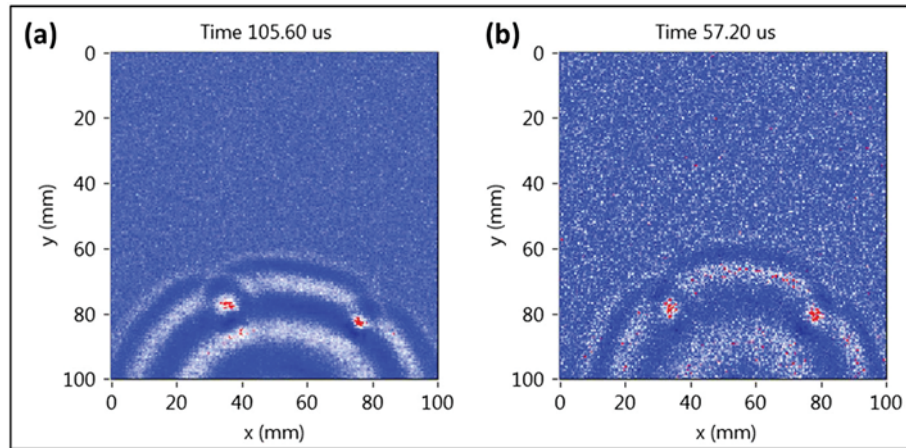


Figure 12: Ultrasonic wave propagation imaging of the carbon fibre-reinforced polymer wing box specimen using (a) capacitive air-coupled transducer, five times repeat scanning, and (b) laser Doppler vibrometer, 15 times repeat scanning

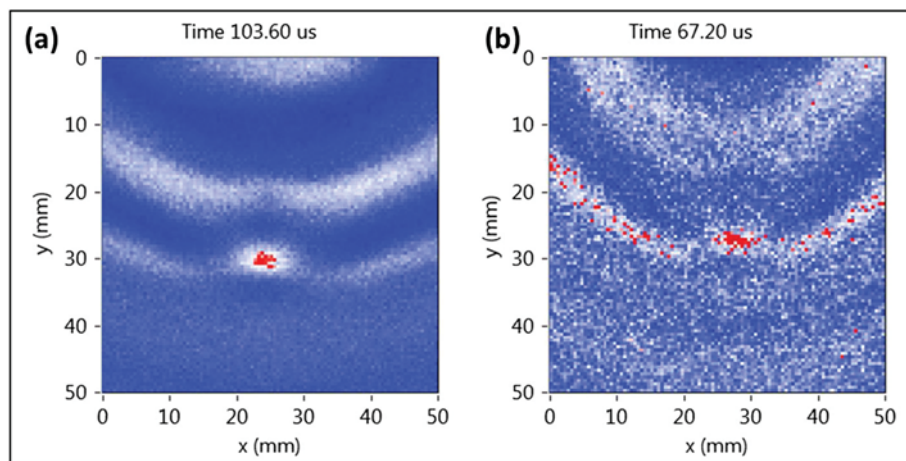


Figure 13: Ultrasonic wave propagation imaging of the scanning test on the 3D-printed object using (a) capacitive air-coupled transducer, five times repeat scanning, and (b) laser Doppler vibrometer, 15 times repeat scanning

Ultrasonic wavenumber imaging algorithm: flowchart

We start with the main idea of the UWI algorithm. As a well-known property, the Lamb waves propagate across a thin plate at various modes, including symmetric modes S_i ($i=0, 1, 2, \dots$) and anti-symmetric modes A_i ($i=0, 1, 2, \dots$) [24]. In the case when the plate has uniform stiffness, at one particular frequency of one particular mode, the guided waves have one corresponding dominant wavenumber. When the waves propagate across a damaged area, which has lower stiffness, the dominant wavenumber at that damaged area alters, such as increasing in the case of the A_0 mode and decreasing in the case of the S_0 mode. By mapping the dominant wavenumber at one particular frequency, it is able to detect the damages as well as to estimate the damage size. The latter capability is unique among damage detection algorithms that have been proposed so far. A brief mathematical development of the

wavenumber imaging algorithm will be presented as follows.

Figure 14 shows the flowchart of the UWI algorithm. To start, the three-dimensional full-wavefield matrix $u(x, y, t)$ is transformed from the space-time domain to the wavenumber-frequency domain using the three-dimensional Fourier transform:

$$U(k_x, k_y, f) = F_{3D}[u(x, y, t)] \quad (5)$$

Next, a narrow frequency bandpass filter centred at frequency f_0 is applied:

$$U_{f_0}(k_x, k_y, f) = U(k_x, k_y, f) W_F(f_0) \quad (6)$$

where $W_F(f_0)$ is a Gaussian-shaped frequency domain window centred at frequency f_0 .

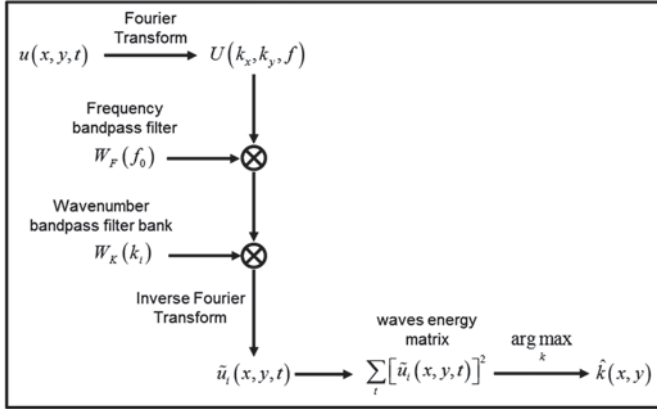


Figure 14: Flowchart of the proposed ultrasonic wavenumber imaging algorithm

Next, the dominant wavenumber at one particular mode of every point in the scan area is calculated. The underlying assumption is that the damage size is relatively small as compared to the whole scan area; therefore, the narrowband frequency waves has a dominant wavenumber k_0 , except at the damaged areas. A 'bank' of wavenumber window $W_K(k_i)$ is built and applied:

$$U_{f_0, k_i}(k_x, k_y, f) = U_{f_0}(k_x, k_y, f) W_K(k_i) \quad (7)$$

where $W_K(k_i)$ is a Gaussian-shaped wavenumber domain window centred at the wavenumber k_i . It is noted that since multiple modes may appear, the range of the 'bank' should be carefully chosen so that it contains only a single mode. We will return to this point with a specific example in the next section.

The result of Equation [7] is a three-dimensional wavefield in the wavenumber–frequency domain with narrowband frequency centred at f_0 and narrowband wavenumber centred at k_i . An inverse Fourier transform is then applied, and the wave energy matrix E_i is calculated as shown in Equations [8] and [9]:

$$\tilde{u}_i(x, y, t) = F_{3D}^{-1}[U_{f_0, k_i}(k_x, k_y, f)] \quad (8)$$

$$E_i(x, y) = \sum_t [\tilde{u}_i(x, y, t)]^2 \quad (9)$$

$E_i(x, y)$ is a two-dimensional matrix with the size of $V \times H$, where V and H are the number scanning points in vertical and horizontal directions. The dominant wavenumber $\hat{k}(x, y)$ at the spatial point (x, y) is the wavenumber that maximises the wave energy at that particular point of wave energy matrices:

$$\hat{k}(x, y) = \underset{k}{\operatorname{argmax}} [E_1(x, y), E_2(x, y), \dots, E_i(x, y) \dots] \quad (10)$$

Ultrasonic wavenumber imaging algorithm: efficient implementation

Since the UWI algorithm is computationally intensive, it is worthy to study an efficient implementation to speed up the algorithm. From a user perspective, the best scenario would be to have an extremely efficient implementation that is able to compute the wavenumber map almost in real time.

The first efficient implementation is to omit the step of mode isolation. In the wavenumber processing, the mode isolation is generally done by estimating the wavenumber–frequency curve and then applying a frequency-dependent bandpass filter in the wavenumber domain. However, since the UWI algorithm only considers a narrowband frequency to map the dominant wavenumber, the mode isolation step can be simply done by selecting an appropriate wavenumber range so that it contains only one mode. An example is shown using the full-wavefield data of the CFRP wing box specimen measured using the ACT sensor. Figure 15a shows the wavenumber–frequency curve, and Figure 15b shows the wavenumber spectrum at the frequency of 130 kHz [25]. At the frequency of 130 kHz, two modes are co-existing (A_0 and S_0) where the A_0 mode has the shorter wavelength and therefore higher wavenumber. The dominant wavenumber of A_0 mode at 130 kHz can be seen from Figure 15b and is around the value of 0.14 mm^{-1} . Therefore, in order to select only the A_0 mode, the appropriate range of the 'bank' of wavenumber window can be chosen such as from 0.07 to 0.2 mm^{-1} . The lower range of the wavenumber window is to avoid the selection of the S_0 mode, which has the dominant wavenumber around the value of 0.04 mm^{-1} . The code to estimate that wavenumber range is quite straightforward. For example, one can easily find two 'radii' which contain the most magnitude in the plane of wavenumber spectrum and then assign the larger value among them to the A_0 mode. It is also possible to extend the case to when there are more than two modes co-existing.

The second efficient implementation is to optimise the coupled Equations [7] and [8] using the Parseval theorem [26]:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \quad (11)$$

where $X[k]$ is the discrete Fourier transform of $x[n]$, both of length N . We can easily realise that Equations [8] and [9] can be replaced by

$$v_i(x, y, f) = F_{2D}^{-1}[U_{f_0, k_i}(k_x, k_y, f)] \quad (12)$$

$$E_i(x, y) = \frac{1}{N^2} \sum_f [v_i(x, y, f)]^2 \quad (13)$$

where F_{2D}^{-1} is the two-dimensional inverse Fourier transform.

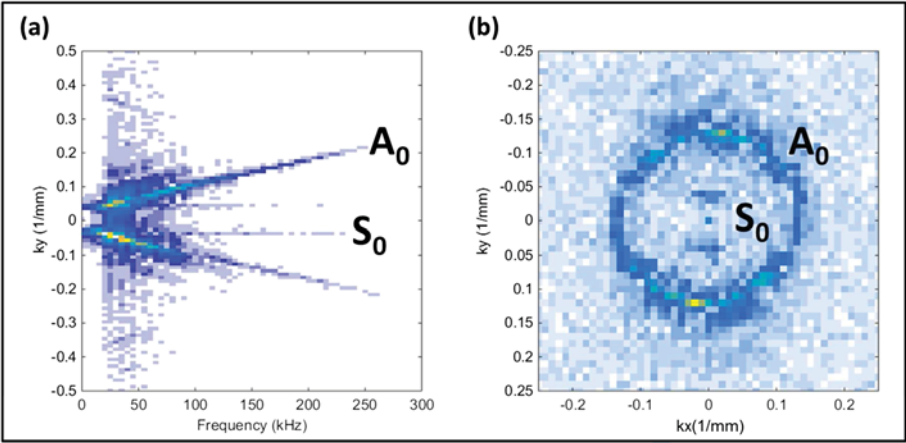


Figure 15: Frequency–wavenumber analysis of full wavefield data of the carbon fibre-reinforced polymer plate (a) wavenumber–frequency curve and (b) wavenumber spectrum at a frequency of 130 kHz

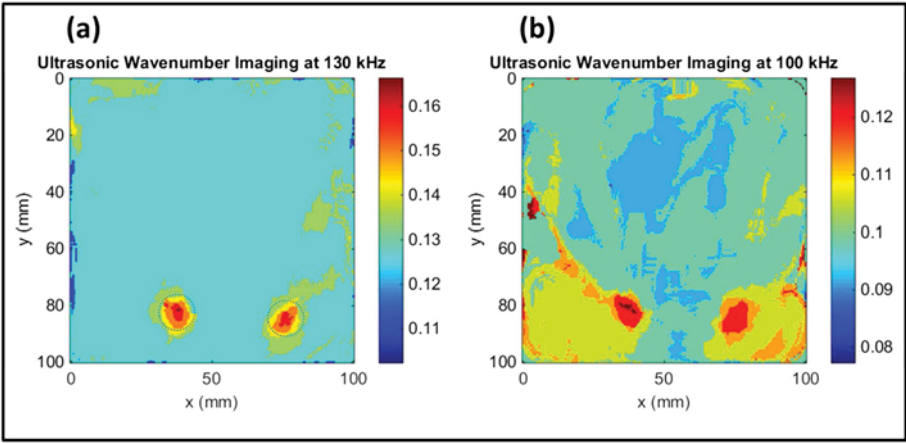


Figure 16: Ultrasonic wavenumber imaging (A_0 mode) of the carbon fibre-reinforced polymer wing box specimen using (a) the capacitive air-coupled transducer and (b) the laser Doppler vibrometer

Another ongoing research approach concerns the use of dedicated hardware, such as the Field-Programmable Gate Array for Fourier transform computation [27]. Since the Fourier transform, including both forward and inverse of the UWI algorithm, took approximately 70% of total computational time, this is a very promising approach towards a real-time algorithm.

Ultrasonic wavenumber imaging algorithm: Results and comparison

Figure 16a and b shows the UWI results for the CFRP wing box specimen using the capacitive ACT and LDV, respectively. The damage detection performance of the ACT is clearly superior. The UWI map in the case of using the LDV sensor has a wavenumber that varies significantly even in the undamaged area. This problem is due to the low SNR of the LDV sensing signals. Readers can find the same conclusion in a recent study of Flynn *et al.* [28] where

Table 1: Comparison between laser Doppler vibrometer (LDV) and air-coupled transducer (ACT) systems for Ultrasonic Propagation Imager

	Advantages	Disadvantages
LDV	Extraordinarily long range (metres) Easy to set up, the laser beam is optimised normal to the surface	Lower signal-to-noise ratio (SNR) as compared to ACT Often needs surface treatment High cost
ACT	Higher SNR than LDV No surface treatment needed Lower cost	Short range (centimetres) Relatively complicated set-up since the optimum measurement angle depends on each specimen

the UWI map using LDV sensor is poorer than in the case when using a highly sensitive affixed transducer.

The damage size estimated from the UWI map is approximately 20 mm in diameter. This is confirmed by using the C-scan image acquired by the Ez-scan VII system developed by Orient NDT Inc. (Goyang city, Republic of Korea) where the damages areas were identified at 16-mm diameter.

Conclusions

We investigated two non-contact ultrasonic sensors, the LDV and the capacitive ACT, for use in the UPI system. The working principle of each sensor was studied to optimise their measurement parameters. Two experiments for damage detection were presented and showed that both the LDV and the capacitive ACT have the capability to serve as practical sensing systems for the UPI. However, each sensor has its own advantages and disadvantages as summarised in Table 1. For the LDV, the most important advantage is its extraordinary long-range measurement. However, the high cost and the need of surface treatment seriously obstructed this technique to a wider user. Without a special need of far-distance sensing, the capacitive ACT with lower cost is generally recommended for non-contact sensing.

Another contribution of this paper is to reintroduce and reinvent the UWI algorithm. For the purpose of performance comparison, in this paper, we reintroduced the UWI algorithm with the intention of presenting it in an approachable manner. The absence of this algorithm in the latest review [3] of full-wavefield signal processing shows there is a need for that. Some new development of the UWI algorithm towards an efficient implementation was presented. The uniqueness of the proposed algorithm with the capability of damage size estimation makes this algorithm very attractive for the future study with full-wavefield signal processing.

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References

1. Raghavan, A., and Cesnik, C. E. (2007) Review of guided-wave structural health monitoring. *Shock. Vib. Dig.* **39**, 91–114.
2. Worden, K., Farrar, C. R., Manson, G., and Park, G. (2007) The fundamental axioms of structural health monitoring. *Proc. R. Soc. A.* **463**, 1639–1664.
3. Ostachowicz, W., Radzienski, M., and Kudela, P. (2014) Comparison studies of full wavefield signal processing for crack detection. *Strain* **50**, 275–291.
4. Michaels, T. E., Michaels, J. E., and Ruzzene, M. (2011) Frequency–wavenumber domain analysis of guided wave fields. *Ultrasonics* **51**, 452–466.
5. Staszewski, W. J., Lee, B. C., Mallet, L., et al. (2004) Structural health monitoring using scanning laser vibrometry: I. Lamb wave sensing. *Smart. Mater. Struct.* **13**, 251–260.
6. Lee, J. R., Takatsubo, J., and Toyama, N. (2007) Disbond monitoring at wing stringer tip based on built-in ultrasonic transducers and a pulsed laser. *Smart. Mater. Struct.* **16**, 1025–1035.
7. Chong, S. Y. and Lee, J. R. (2014) Development of laser ultrasonic propagation imaging system with twenty-kilohertz scanning frequency for nondestructive evaluation applications. *P. of Advances in Structural Health Management and Composite Structures*, Vol I, 181–184.
8. Shin, H. J. and Lee, J. R. (2013) Development of smart hangar technology based on a long range laser ultrasonic propagation imaging system. *P. of Korea Society of Nondestructive Testing*.
9. Scruby, C. B., and Drain, L. E. (1990) *Laser Ultrasonics: Techniques and Applications*. CRC Press, New York.
10. Cosenza, C. (2007) Generation of narrowband antisymmetric lamb waves using a formed laser source in the ablative regime. *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **54**, 147–156.
11. Huang, J. (1993) Crack detection in fuselage panels by a narrow-band laser-based ultrasonic system. *Rev. Prog. Quant. Nondestr. Eval.* **12**, 603–610.
12. Scalea, F. L. (1999) Remote laser generation of narrow-band surface waves through optical fibers. *IEEE Trans. Ultrason., Ferroelectr., Freq. Control.* **46**, 1551–1557.
13. Nishino, H. (1993) Excitation of high frequency surface acoustic waves by phase velocity scanning of a laser interference fringe. *Appl. Phys. Lett.* **62**, 2036–2038.
14. Pierce, R., Ume, C., and Jarzynski, J. (1995) Temporal modulation of a laser source for the generation of ultrasonic waves. *Ultrasonics* **33**, 133–137.
15. Wagner, J. W., Deaton, J. B., and Spicer, J. B. (1988) Generation of ultrasound by repetitively Q-switching a pulsed Nd:YAG laser. *Opt. Soc. Am.* **27**, 4696–4700.
16. Pierce, S. (1998) Laser generation of ultrasound using a modulated continuous wave laser diode. *Appl. Phys. Lett.* **27**, 1030–1032.
17. Nakamura, K. (2012) *Ultrasonic Transducers Materials and Design for Sensors, Actuators and Medical Applications*. Woodhead Publishing Limited, Cambridge: 374–407.
18. *Laser Doppler vibrometer manual of Polytec, Inc.*
19. Lee, J. R., Chong, S. Y., Sunuwar, N., et al. (2013) Repeat scanning technology for laser ultrasonic propagation imaging. *Meas. Sci. Technol.* **24**, 085201.
20. Chia, C. C., Lee, J. R., Park, C. Y., and Jeong, H. M. (2012) Laser ultrasonic anomalous wave propagation imaging method with adjacent wave subtraction application to actual damages in composite wing. *Opt. Laser. Technol.* **44**, 428–440.
21. Flynn, E. B., Chong, S. Y., Jarmer, G. J., et al. (2013) Structural imaging through local wavenumber estimation of guided waves. *NDT&E Int.* **59**, 1–10.
22. Rogge, M. D., and Leckey, C. A. C. (2013) Characterization of impact damage in composite laminates using guided wavefield imaging and local wavenumber domain analysis. *Ultrasonics* **53**, 1217–1226.

23. Mesnil, M., Leckey, C. A. C., and Ruzzene, M. (2014) Instantaneous and local wavenumber estimations for damage quantification in composites. *Struct. Health. Monit.* **1**, 193–204.
24. Rose, J. (2004) *Ultrasonic Waves in Solid Media*. Cambridge University Press, Cambridge; New York.
25. Tian, Z., and Yu, L. (2014) Lamb wave frequency–wavenumber analysis and decomposition. *J. Intel. Mat. Syst. Str.* **1**, 1107–1123.
26. Alan V. and Schafer R. W. (1999) *Discrete-time Signal Processing*, 2nd Edition. Prentice Hall: Upper Saddle River, NJ.
27. Kestur, S., Park, S., Irick, K. M. and Narayanan, V. (2010) Accelerating the nonuniform fast Fourier transform using FPGAs, *18th IEEE Annual International Symposium on Field-Programmable Custom Computing Machines*, 19–26.
28. Flynn, E. B. and Chong, S. Y. (2014) Acoustic wavenumber spectroscopy for fully non-contact ultrasonic pulse imaging. *P. of Advances in Structural Health Management and Composite Structures*, Vol I, 61–64.